

CH 101: General Chemistry I

Fall 2026 · 4 credit hours · Sample syllabus for the CourseCompass demo (synthetic)

Grading

Component	Weight	Due
Midterm exam	25%	Week 7
Homework sets 1-6	20%	Weekly
Quizzes 1-4	10%	Weeks 3, 5, 9, 11
Lab reports	20%	After each lab
Final exam (cumulative)	25%	Finals week

Unit 1: Atoms, moles and stoichiometry

A mole is 6.022×10^{23} particles (Avogadro's number). Molar mass, in grams per mole, converts between the mass of a substance and the number of moles. In a balanced chemical equation the coefficients give mole ratios between reactants and products. The limiting reactant is the one that runs out first and determines the theoretical yield; percent yield equals actual yield divided by theoretical yield, times 100.

Unit 2: Gases

The ideal gas law is $PV = nRT$, where $R = 0.0821 \text{ L atm per mol K}$. Temperature must be in kelvin ($K = \text{degrees C} + 273.15$). Boyle's law: at constant temperature, pressure and volume are inversely related. Charles's law: at constant pressure, volume is proportional to absolute temperature. Dalton's law: the total pressure of a gas mixture is the sum of the partial pressures.

Unit 3: Thermochemistry

Enthalpy change (ΔH) is negative for exothermic reactions, which release heat, and positive for endothermic reactions, which absorb heat. Heat $q = m c \Delta T$, where c is specific heat capacity ($4.184 \text{ J per g per degree C}$ for water). Hess's law: the enthalpy change of a reaction is the same whether it happens in one step or several, so step enthalpies can be added.

Unit 4: Atomic structure and bonding

Electrons fill orbitals in order of increasing energy (aufbau principle). Electronegativity increases across a period and up a group; fluorine is the most electronegative element. Ionic bonds form between metals and nonmetals by electron transfer; covalent bonds share electrons. VSEPR theory predicts molecular shape from electron pairs around the central atom: two pairs give a linear shape, four give a tetrahedral shape (109.5 degrees).

Policies

Late homework loses 10% per day. Office hours: Tuesdays and Thursdays, 2-4 pm. The last day to withdraw with a W is the end of week 10. Talk to your academic advisor before withdrawing.

